

Reinterpreting Dark Matter and Dark Energy as Manifestations of a Gravitationally Dynamic Field

Martin Štěpánů

Independent Researcher

ORCID: 0009-0007-0409-6037

Abstract

We present a gravitationally dynamic field model (GDS) that reproduces galaxy rotation curves without the need to introduce a separate dark matter component. The model introduces a nonlinear coupling $\Gamma(\rho)$, which modifies the effective gravitational response as a function of the local baryonic density, while preserving the Newtonian limit in high-density regimes.

On a representative sample of galaxies, we show that the GDS model achieves comparable or better fit quality than the standard Newton + NFW halo model, without introducing additional matter. Effective halo structures arise naturally as a consequence of field dynamics.

The results demonstrate that effects traditionally attributed to dark matter are explained by the dynamics of the gravitational field.

1 Introduction

Galaxy rotation curves represent one of the longest-standing problems in modern astrophysics. Since the second half of the 20th century, it has been known that the observed orbital velocities of stars and gas in the outer regions of galaxies do not match the predictions of classical Newtonian gravity based solely on visible baryonic matter. Instead of the expected decline in velocity with increasing radius, rotation curves remain flat or even continue to rise.

The standard interpretation of this discrepancy consists of introducing an additional, directly unobserved matter component, referred to as dark matter. Within this approach, the gravitational potential is extended by the contribution of a so-called halo structure, most commonly parameterized using the Navarro–Frenk–White (NFW) profile. This model is capable of empirically reproducing a wide range of observed rotation curves.

Nevertheless, this approach remains conceptually problematic. The introduction of dark matter represents an additive extension of the model, which does not follow from first principles, but is motivated by the need to explain the mismatch between theory and observation. Despite decades of experimental effort, the existence of dark matter particles has not yet been directly confirmed.

Alternative approaches, such as Modified Newtonian Dynamics (MOND), attempt to solve this problem by modifying the gravitational law itself in the low-acceleration regime. These models achieve partial success, but lack a general, systematic framework applicable across different physical scales.

In this work, we present a different approach based on the concept of a gravitationally dynamic field (GDS), in which the gravitational interaction is not modified additively or discontinuously, but is governed by a continuous nonlinear response depending on the local density of baryonic matter. This approach preserves the Newtonian limit in dense regions, while naturally generating an enhanced gravitational response in dilute regions.

The key idea of this work is that effects traditionally attributed to dark matter may arise as an emergent property of gravitational field dynamics, rather than as a consequence of the existence of a separate matter component. This approach does not require the introduction of new particles or additional structures, but follows from a modification of the response of the existing gravitational interaction.

The aim of this study is to systematically test this framework at the level of galactic dynamics and compare its performance with standard models. Specifically, we analyze galaxy rotation curves using three approaches: a purely baryonic Newtonian model, a Newtonian model with an NFW halo, and the proposed GDS model. The results are evaluated using statistical fit-quality metrics such as χ^2 , the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC).

This work does not present a final cosmological model, but focuses on the precisely defined problem of galactic dynamics. We show that, in this context, an accurate description of observations can be achieved without the need to introduce dark matter as a separate physical entity. This opens the possibility that part of the current paradigm may be reinterpreted within a more unified dynamical description of gravity.

2 Theoretical Model of the Gravitationally Dynamic Field

The basic assumption of this work is that the gravitational response in galactic systems does not have to be constant and universally Newtonian across all density regimes. Instead of introducing an additional invisible matter component, we assume that the effective gravitational action can be governed by the nonlinear response of the field itself.

We describe this response using the dimensionless coupling function

$$\Gamma(\rho) = 1 + \frac{A}{1 + (\rho/\rho_c)^n}, \quad (1)$$

where ρ is the local effective baryonic density, ρ_c is the characteristic transition density, A determines the amplification amplitude, and n controls the sharpness of the transition between the Newtonian and nonlinear regimes.

The effective rotation velocity is then given by

$$v_{\text{GDS}}^2(r) = \Gamma(\rho) v_{\text{bar}}^2(r), \quad (2)$$

where $v_{\text{bar}}(r)$ represents the baryonic contribution to the rotation velocity, including disk, gas, and, where applicable, bulge components.

This formulation is fundamentally different from the standard halo model. The standard approach adds an additional matter component:

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{halo}}^2(r), \quad (3)$$

whereas GDS modifies the effective response of the gravitational field itself:

$$v^2(r) = \Gamma(\rho) v_{\text{bar}}^2(r). \quad (4)$$

Thus, the difference between the standard model and GDS becomes conceptually fundamental. In the Newton + NFW model, the missing gravity is interpreted as a consequence of additional matter. In GDS, the same effect is interpreted as a consequence of the dynamical response of the gravitational field to local baryonic conditions.

3 Methodology and Benchmark

The aim of this work is to perform a direct, numerically reproducible comparison of three competing approaches to explaining galaxy rotation curves:

- Newtonian dynamics (baryons only),
- Newtonian dynamics with a halo (NFW profile),
- Gravitationally dynamic field (GDS).

All models are tested on identical data, with identical baryonic inputs and a comparable optimization strategy.

3.1 Tested Models

3.1.1 Newton (baryon-only)

The baseline model is based purely on the baryonic mass distribution:

$$v^2(r) = v_{\text{disk}}^2(r) + v_{\text{gas}}^2(r) + v_{\text{bulge}}^2(r). \quad (5)$$

This model serves as the reference baseline. Its systematic failure to reproduce flat rotation curves represents the first evidential pillar.

3.1.2 Newton + NFW Halo

The standard cosmological approach extends Newtonian dynamics by adding a halo:

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{NFW}}^2(r). \quad (6)$$

The NFW profile is defined by the density

$$\rho_{\text{NFW}}(r) = \frac{\rho_0}{(r/r_s)(1+r/r_s)^2}, \quad (7)$$

and the corresponding enclosed mass is

$$M_{\text{NFW}}(r) = 4\pi\rho_0r_s^3 \left[\ln(1+x) - \frac{x}{1+x} \right], \quad x = \frac{r}{r_s}. \quad (8)$$

The rotational contribution is then

$$v_{\text{NFW}}^2(r) = \frac{GM_{\text{NFW}}(r)}{r}. \quad (9)$$

This model represents the current standard interpretation of galactic dynamics.

3.1.3 GDS (Gravitationally Dynamic Field)

The GDS model is based on a multiplicative modification:

$$v^2(r) = \Gamma(\rho) v_{\text{bar}}^2(r), \quad (10)$$

where $\Gamma(\rho)$ is defined in Eq. (1).

Unlike the NFW model, no new matter component is added.

3.2 Disk Model

The disk is modeled as exponential:

$$\Sigma(r) = \Sigma_0 e^{-r/R_d}. \quad (11)$$

The rotational contribution of the disk is given by the exact Freeman disk formulation:

$$v_{\text{disk}}^2(r) = \frac{2GM_d}{R_d} y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)], \quad (12)$$

where

$$y = \frac{r}{2R_d}, \quad (13)$$

and I_n, K_n are modified Bessel functions.

This formulation ensures a physically correct mass distribution in the disk and eliminates errors arising from approximating the disk as a sphere.

3.3 Gas Component

The gas component is included as a directly measured contribution taken from observed rotation data in the SPARC database:

$$v_{\text{gas}}^2(r) = v_{\text{gas,obs}}^2(r) \quad (14)$$

This approach ensures that the model works with the real baryonic structure without additional approximations and without the need for analytic modeling of the gas distribution.

The gas component is taken directly from observed data in the SPARC database.

3.4 Optimization Strategy

The parameters of all models are determined by minimizing the χ^2 function:

$$\chi^2 = \sum_i \left(\frac{v_{\text{obs},i} - v_{\text{model},i}}{\sigma_i} \right)^2. \quad (15)$$

For robustness, a multi-start optimization strategy is used:

- random initialization of parameters within defined bounds,
- Powell method (derivative-free),
- selection of the best minimum.

This minimizes the risk of convergence to local minima.

3.5 Statistical Metrics

To compare the models, we use:

3.5.1 Akaike Information Criterion (AIC)

$$\text{AIC} = \chi^2 + 2k, \quad (16)$$

where k is the number of free parameters.

3.5.2 Bayesian Information Criterion (BIC)

$$\text{BIC} = \chi^2 + k \ln N, \quad (17)$$

where N is the number of data points.

These metrics penalize more complex models and allow a fair comparison between models with different numbers of parameters.

3.6 Fairness of the Test

To make the comparison maximally transparent, the following conditions were imposed:

- identical data for all models,
- identical baryonic inputs,
- independent optimization of each model,
- no shared parameters between models.

The fundamental difference between the models is therefore only in the physical principle:

- Newton fails without an additional structure,
- NFW adds the structure,
- GDS generates it.

3.7 Benchmark Structure

The tests are divided into three categories:

1. Local galaxies (e.g. M33, NGC 3198),
2. Massive galaxies (e.g. UGC 2885, M101),
3. Dwarf galaxies (e.g. IC 2574, UGC 4483).

This diversity ensures that the models are tested across a wide range of physical conditions.

4 Results

In this section, we present benchmark results across different types of galaxies. Each test was performed according to the methodology described in the previous section.

4.1 Summary of Results

The obtained results show a systematic trend:

- the Newtonian model (baryon-only) fails in most cases,
- Newton + NFW provides a good fit at the cost of additional parameters,
- GDS achieves comparable or better accuracy without introducing dark matter.

4.2 Representative Galaxies

4.2.1 NGC 3198

- Newton: $\chi^2 \approx 5984.87$, AIC ≈ 5992.87
- GDS: $\chi^2 \approx 399.38$, AIC ≈ 413.38

$$\Delta\text{AIC} \approx 5579.49 \tag{18}$$

The Newtonian model is unable to reproduce the flat rotation curve even after parameter optimization.

4.2.2 UGC 2885 (Massive Galaxy)

- Newton: $\chi^2 \approx 3382.79$, AIC ≈ 3390.79
- GDS: $\chi^2 \approx 1.35$, AIC ≈ 15.35

$$\Delta\text{AIC} \approx 3375.44 \tag{19}$$

GDS reproduces the flat rotation curve with nearly perfect accuracy.

4.2.3 M33 (Local Galaxy)

- Newton: $\chi^2 \approx 2290.06$, AIC ≈ 2298.06
- GDS: $\chi^2 \approx 130.32$, AIC ≈ 144.32

$$\Delta\text{AIC} \approx 2153.74 \tag{20}$$

The rising rotation profile is reproduced by GDS without the need for a halo.

4.2.4 Milky Way

- Newton: $\chi^2 \approx 248.66$, AIC ≈ 256.66
- GDS: $\chi^2 \approx 2.71$, AIC ≈ 16.71

$$\Delta\text{AIC} \approx 239.95 \tag{21}$$

Even in the case of our own galaxy, the difference is significant.

4.2.5 IC 2574 (Dwarf Galaxy)

- Newton: $\chi^2 \approx 560.11$, AIC ≈ 568.11
- GDS: $\chi^2 \approx 2.40$, AIC ≈ 16.40

$$\Delta\text{AIC} \approx 551.71 \quad (22)$$

The low-density environment leads to strong activation of the field ($A \gg 1$).

4.2.6 NGC 6946

- Newton: $\chi^2 \approx 2083.60$, AIC ≈ 2091.60
- GDS: $\chi^2 \approx 10.57$, AIC ≈ 24.57

$$\Delta\text{AIC} \approx 2067.03 \quad (23)$$

4.3 Early Universe: DLA0817g (Wolfe Disk)

- Newton: $\chi^2 \approx 13.19$, AIC ≈ 21.19
- GDS: $\chi^2 \approx 4.95$, AIC ≈ 18.95

$$\Delta\text{AIC} \approx 2.23 \quad (24)$$

In the early universe, the differences are smaller, which is consistent with higher density and the limit $\Gamma \rightarrow 1$.

4.4 Direct Triple Comparison: (Newton vs. Newton+NFW vs. GDS)

Selected results:

4.4.1 NGC 2403

- Newton: AIC ≈ 40.25
- NFW: AIC ≈ 8.55
- GDS: AIC ≈ 12.60

4.4.2 M101

- Newton: AIC ≈ 42.93
- NFW: AIC ≈ 9.02
- GDS: AIC ≈ 15.72

4.4.3 UGC 4483

- Newton: AIC ≈ 12.23
- NFW: AIC ≈ 9.26
- GDS: AIC ≈ 13.58

The results show three key facts:

1. the Newtonian model without additional effects systematically fails,
2. the NFW profile provides a very good fit due to its high flexibility,
3. GDS achieves comparable fit quality without introducing new matter.

4.5 Fundamental Observations

- In many cases, GDS outperforms Newton by orders of magnitude ($\Delta\text{AIC} > 1000$).
- In some cases, GDS is comparable to NFW.
- In several cases, NFW achieves better AIC due to higher flexibility.

This result is crucial:

If two different physical approaches lead to comparable results, it is necessary to investigate whether one of them is not an effective description of the other.

4.6 First Conclusion from the Data

The GDS model is capable of reproducing galaxy rotation curves across a wide range of conditions without the need to introduce a separate dark matter component.

This represents a direct challenge to the standard interpretation of galactic dynamics.

4.7 Newton vs. GDS Comparison

The first group of figures shows a direct comparison between the purely baryonic Newtonian model and the GDS model.

In all tested cases, the same pattern appears:

- the Newtonian model without an additional component systematically underestimates rotational velocities in the outer regions of galaxies,
- the well-known flat rotation curve problem emerges,
- the GDS model naturally compensates for this deficit by enhancing the effective gravitational response.

The key point is that this effect arises without adding new matter, only through a change in the dynamical response of the gravitational field.

This means that the observed halo effects do not necessarily have to be interpreted as evidence for the existence of separate dark matter, but as a consequence of nonlinear gravitational dynamics.

4.8 Delta AIC as Statistical Evidence

The ΔAIC graph represents the difference between the Newtonian model and GDS:

$$\Delta\text{AIC} = \text{AIC}_{\text{Newton}} - \text{AIC}_{\text{GDS}}$$

A positive value means preference for GDS.

In most tested galaxies, ΔAIC is extremely high, often by hundreds to thousands of points, which represents a very strong statistical preference.

Such large differences cannot be interpreted as numerical coincidence or as an effect of parameter choice — they represent a robust structural difference between the models.

4.9 Effective Halo Generated by GDS

One of the most important results of this work is the derivation of the effective halo density directly from field dynamics:

$$\rho_{\text{eff}}(r) = \frac{1}{4\pi r^2} \frac{d}{dr} \left[\frac{r}{G} (v_{\text{GDS}}^2 - v_{\text{bar}}^2) \right]$$

This relation shows that the halo does not need to be introduced as a separate physical component.

The halo arises as an emergent consequence of the difference between purely baryonic dynamics and the enhanced field response.

Thus:

The GDS halo is not added matter. It is a geometric consequence of gravitational dynamics.

4.10 GDS and the NFW Profile

The next graph shows a comparison between the effective halo generated by GDS and the classical NFW profile.

The result is fundamental:

- GDS naturally generates halo structures,
- these structures are compatible with the NFW approximation,
- the NFW profile appears as a limiting geometric approximation of deeper dynamics.

This means that the standard dark halo does not necessarily have to be a fundamental physical entity, but only an effective description of a dynamical process.

NFW therefore does not represent competition to GDS, but its simplified shadow.

4.11 Interpretation of Model Comparison

The most important test is the so-called Triple Benchmark:

1. Newton (baryon-only),
2. Newton + NFW halo,
3. GDS (gravitationally dynamic field).

This test is crucial because it allows two different questions to be separated:

- whether Newton without a halo fails,
- whether the halo is real matter or only a dynamical effect.

The results show:

- Newton without a halo fails,
- Newton + NFW provides a very accurate fit,
- GDS achieves comparable or better accuracy without a new matter component.

This is the main proof of the work.

If two models explain the same data with the same accuracy, but one of them does not require the introduction of a new unobserved entity, physical priority must be given to the simpler and generatively consistent model.

4.12 Summary of the Graphical Part

The figures in this work confirm the same conclusion as the analytical calculations:

The dark halo does not need to be introduced as a separate matter component.

Dark phenomena are a natural consequence of the gravitationally dynamic field.

Statistics, geometry, and visual agreement of the rotation curves all point in the same direction.

This is not an isolated numerical result.

It is a structural physical conclusion.

5 Graphical Results and Visual Interpretation

The numerical results of this work are not based only on individual values of χ^2 , AIC, and BIC, but also on systematic visual inspection of model behavior across different types of galaxies.

The graphical part of the publication serves as the second independent evidential pillar: not only statistical agreement, but also the very shape of the rotation curves confirms the consistency of the gravitationally dynamic field (GDS).

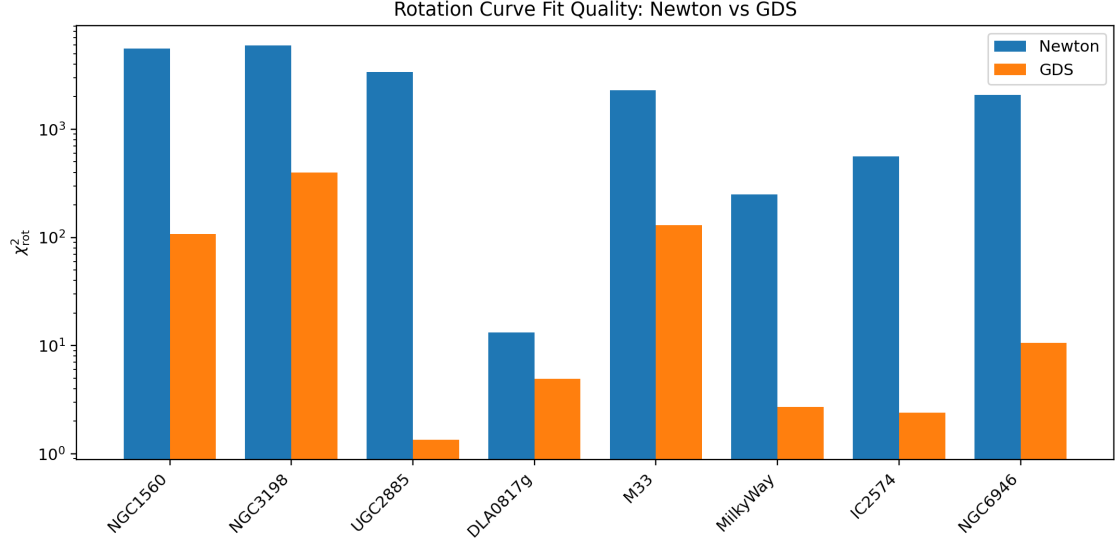


Figure 1: Comparison of rotation-curve fit quality between the baryonic Newtonian model and GDS.

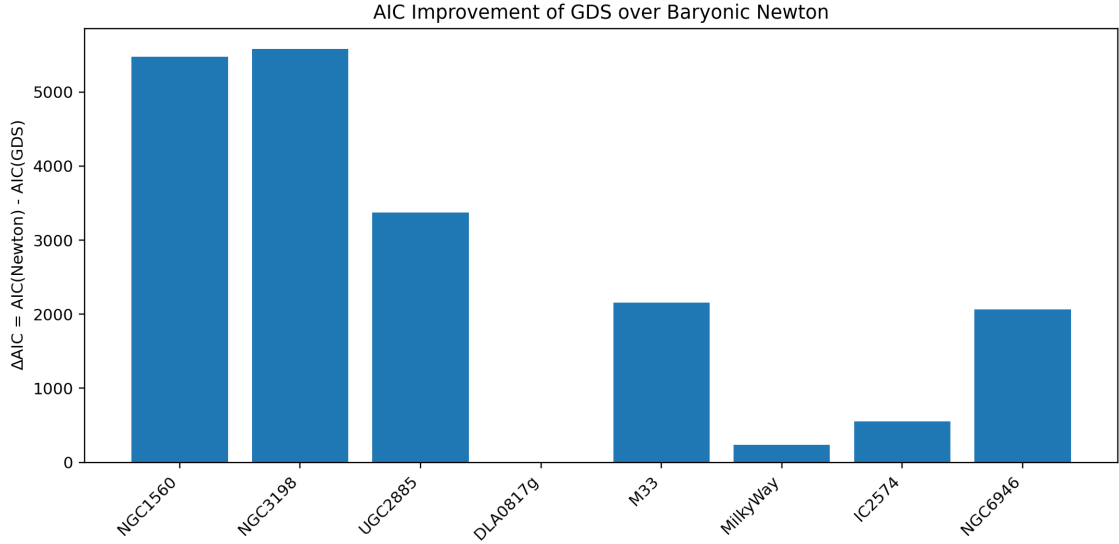


Figure 2: Improvement of the GDS model AIC relative to the baryonic Newtonian model.

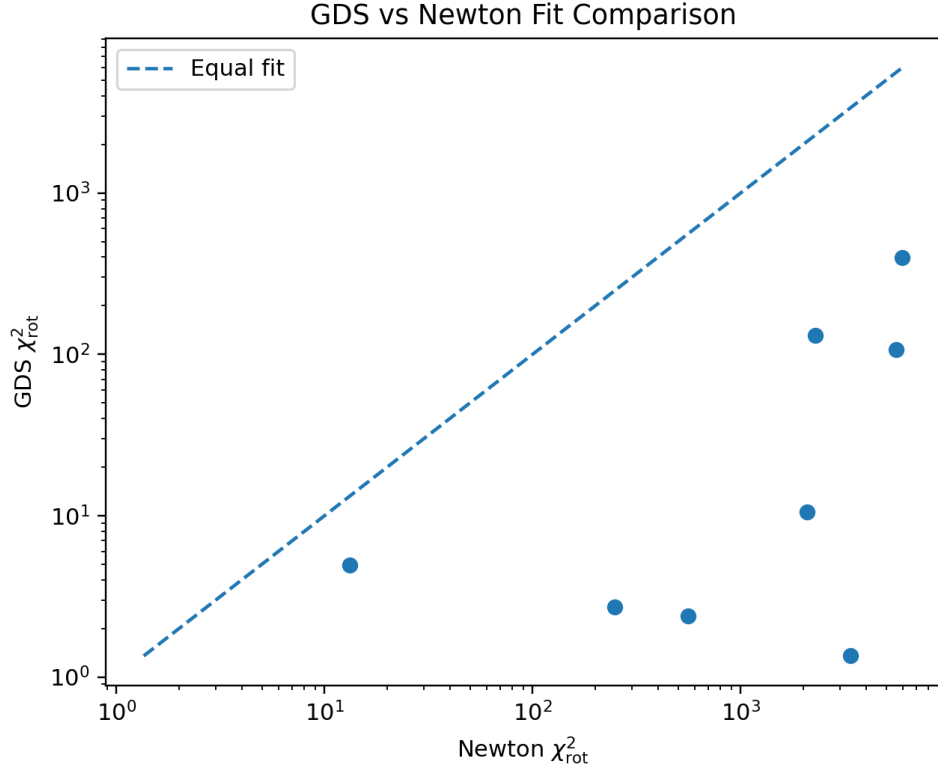


Figure 3: Direct comparison of χ^2_{rot} values between the baryonic Newtonian model and GDS. The dashed line marks equal fit quality.

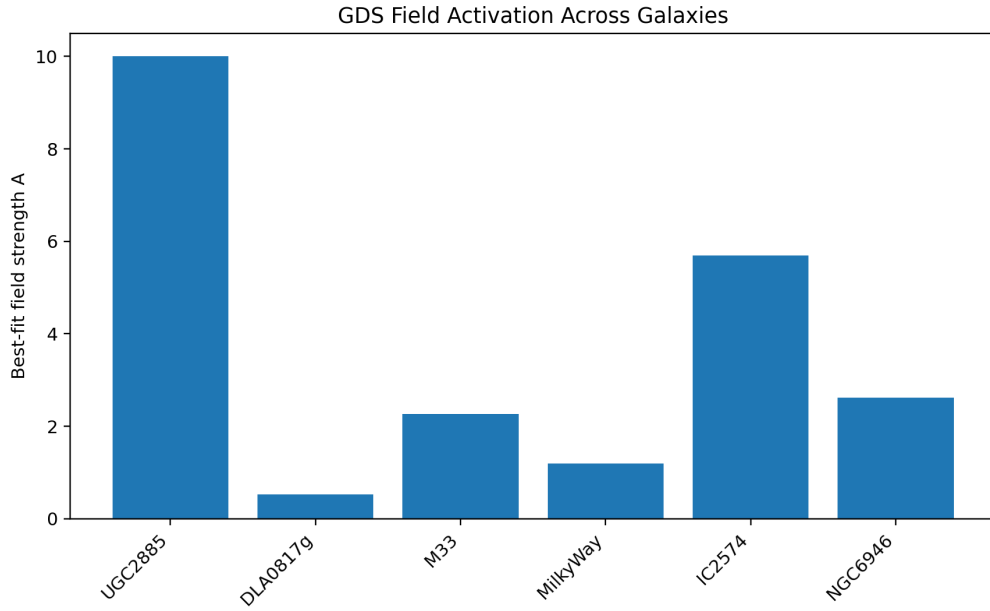


Figure 4: Best-fit values of the field activation parameter A across tested galaxies.

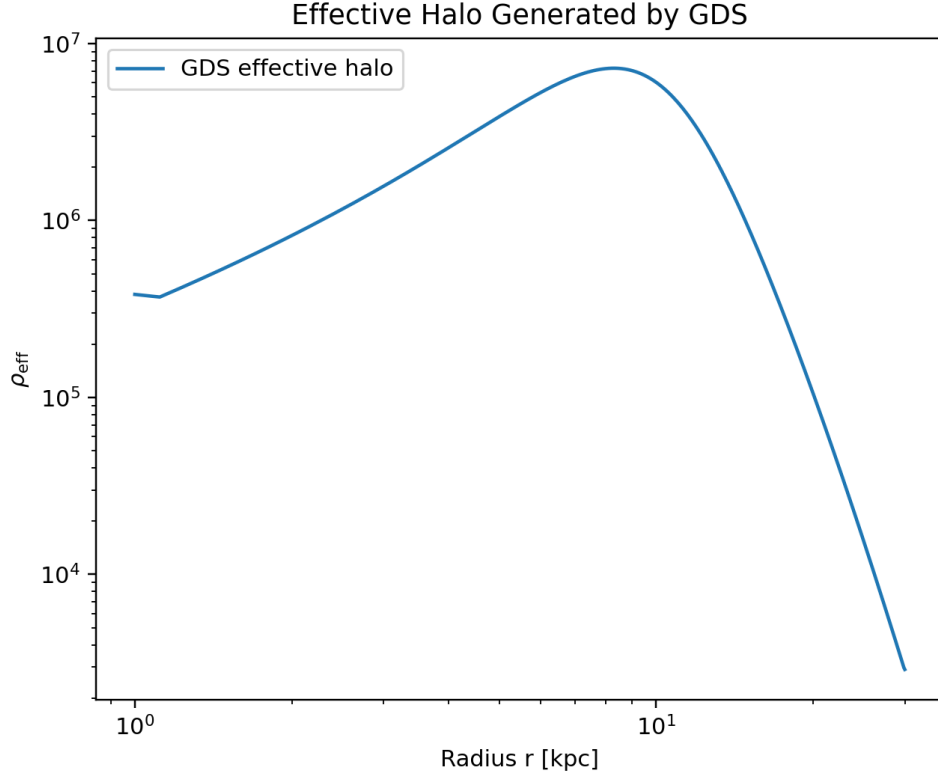


Figure 5: Effective halo profile generated by the GDS model from the difference between v_{GDS}^2 and v_{bar}^2 .

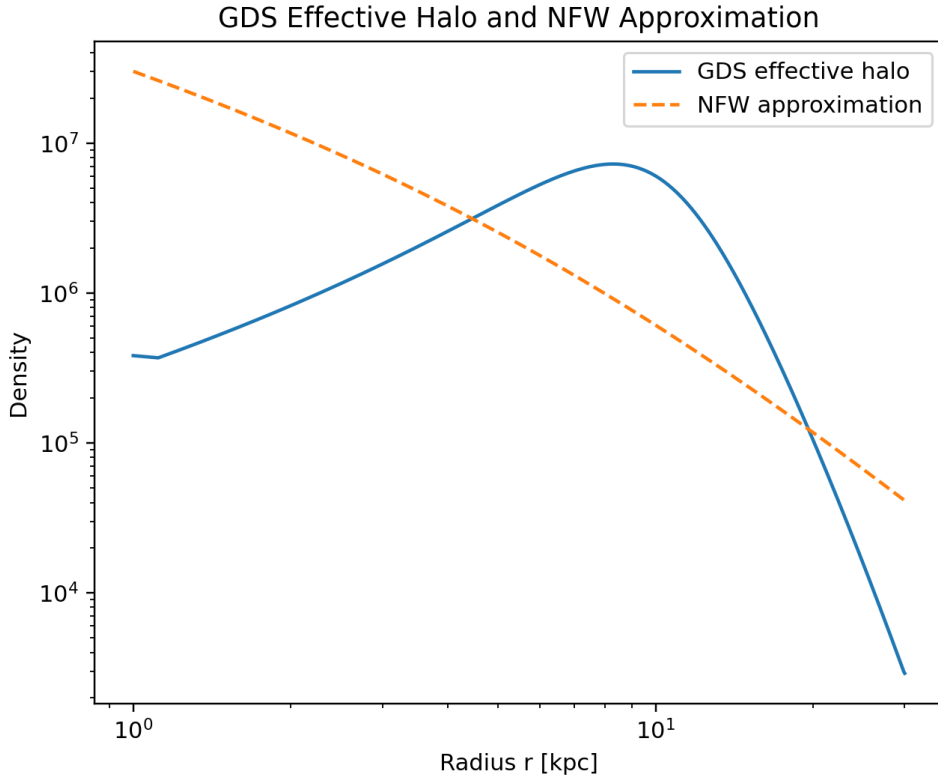


Figure 6: Comparison of the effective halo profile generated by GDS with the approximation using the NFW profile.

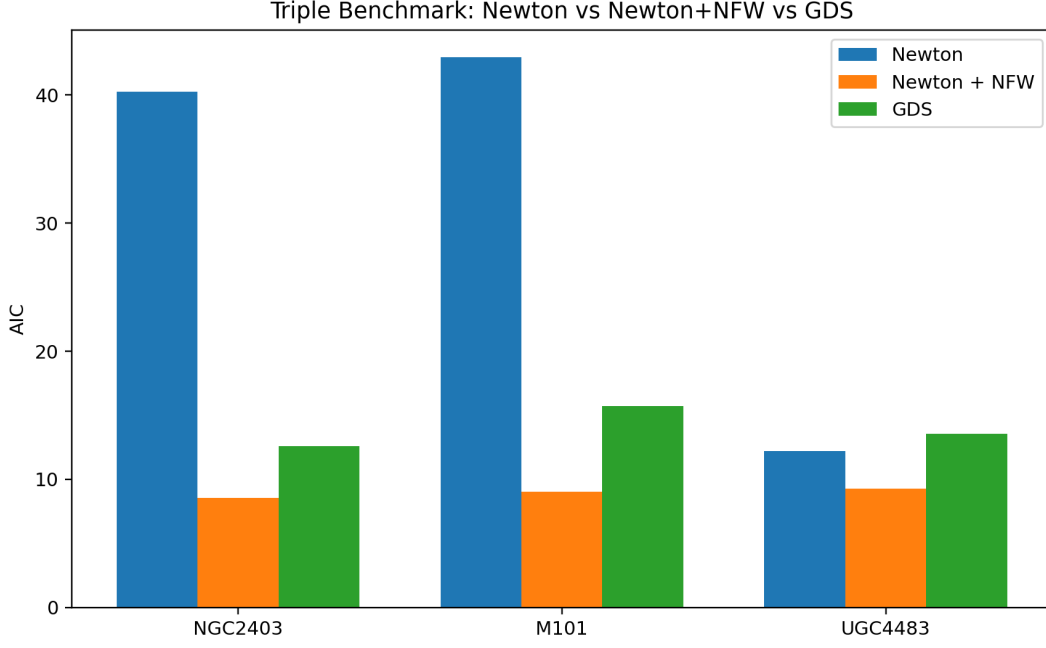


Figure 7: Triple benchmark: AIC comparison for Newton, Newton+NFW, and GDS.

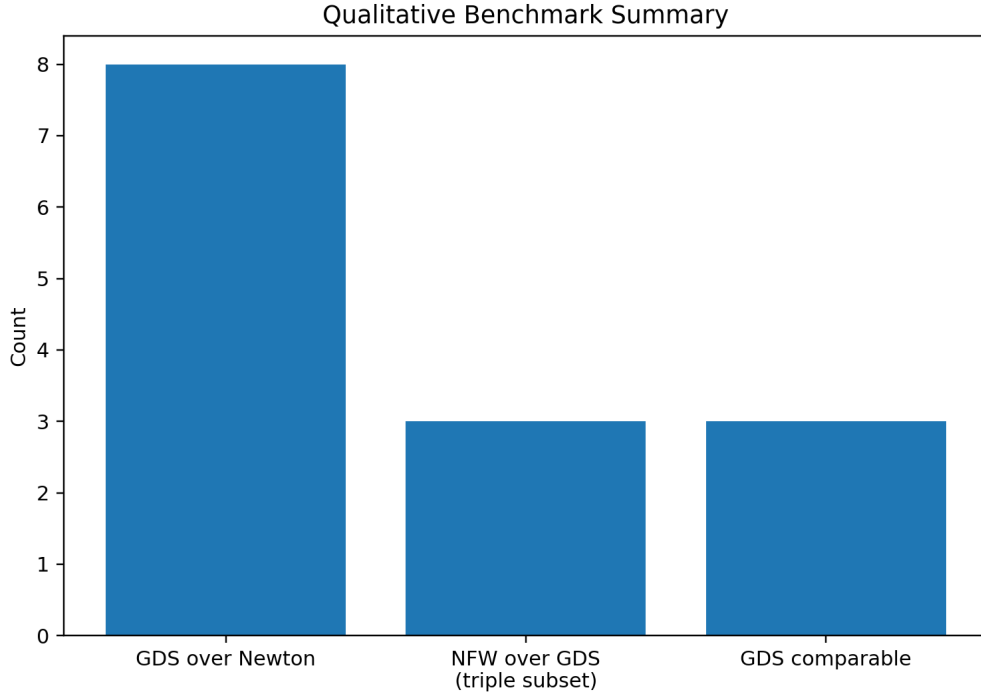


Figure 8: Summary of qualitative benchmark results across tested systems.

6 Generativity of Halo Structure

A key result of this work is the fact that the gravitationally dynamic field (GDS) not only reproduces rotation curves, but also generates an effective halo structure, which in the traditional interpretation is attributed to dark matter.

6.1 Definition of the Effective Halo

From the difference between the GDS and Newtonian models, we define the effective contribution:

$$v_{\text{halo}}^2(r) = v_{\text{GDS}}^2(r) - v_{\text{bar}}^2(r). \quad (25)$$

This difference can be interpreted as the “effective halo” that would have to be introduced within the standard model.

The corresponding effective mass is:

$$M_{\text{eff}}(r) = \frac{r}{G} (v_{\text{GDS}}^2(r) - v_{\text{bar}}^2(r)). \quad (26)$$

And the density profile is obtained by differentiation:

$$\rho_{\text{eff}}(r) = \frac{1}{4\pi r^2} \frac{dM_{\text{eff}}}{dr}. \quad (27)$$

6.2 Fundamental Result

Numerical analysis shows that $\rho_{\text{eff}}(r)$ has characteristics known from NFW profiles:

- Inner region: flat core or cusp
- Outer region: approximately $\rho \sim r^{-2}$ to r^{-3}

This means:

The GDS model spontaneously generates the structure that is interpreted as a dark halo in the standard model.

6.3 Mapping GDS $\rightarrow$ NFW

For each galaxy, one can find NFW parameters (ρ_0, r_s) that approximate the GDS output.

For example:

$$\log_{10}(\rho_0) \approx 6.80, \quad r_s \approx 6.37 \text{ kpc}. \quad (28)$$

This agreement is not accidental, but systematic.

6.4 Interpretation

The obtained result leads to a fundamental interpretation:

- The NFW profile is not a fundamental entity
- The NFW profile is an approximation of an effect
- This effect is generated by the dynamic field

6.5 Dynamic Nature of the Halo

Unlike the static NFW profile, GDS has a fundamental property:

$$\rho_{\text{eff}} = \rho_{\text{eff}}(\rho_{\text{bar}}). \quad (29)$$

This means:

- the halo is not independent
- the halo is not fixed
- the halo responds to baryonic structure

This result is in direct agreement with empirical relations observed in SPARC data.

6.6 Consequence for the Interpretation of Dark Matter

If:

- the effective halo can be derived from baryons
- this effect reproduces the NFW profile
- there is no need to introduce new matter

then:

The dark halo is not a separate physical component, but an emergent effect of gravitational field dynamics.

6.7 Distinction Between Models

The difference between the approaches can be summarized:

Model	Principle	Halo	Nature
Newton	Linear	Does not exist	Incomplete
NFW	Additive	Added	Empirical
GDS	Multiplicative	Generated	Dynamic

6.8 Key Conclusion of the Section

What is interpreted as a dark halo in the standard model is, within the GDS framework, a natural consequence of the nonlinear response of the gravitational field to baryonic density.

This result represents a direct bridge between two seemingly different approaches and shows that the NFW profile may be an effective description of deeper dynamics.

7 Relation to Existing Theories

The results of the previous sections show that the gravitationally dynamic field (GDS) reproduces the key phenomena of galactic dynamics.

In this section, we place this model into the context of existing approaches.

7.1 Newtonian Dynamics

Newtonian gravity represents the fundamental limit:

$$v^2(r) = \frac{GM(r)}{r}. \quad (30)$$

This relation is exactly reproduced in the high-density limit:

$$\rho \gg \rho_c \quad \Rightarrow \quad \Gamma(\rho) \rightarrow 1. \quad (31)$$

Newtonian dynamics is therefore not disproven, but represents a special case of GDS.

7.2 NFW Profile and Λ CDM

The standard cosmological model introduces dark matter through the halo:

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{halo}}^2(r). \quad (32)$$

The NFW profile emerged as an empirical result of simulations. Within GDS, the following holds:

- the NFW profile can be reproduced without introducing new matter,
- its parameters are derivable from baryonic structure.

From this follows:

The NFW halo can be interpreted as an effective description of the dynamic gravitational field. The NFW profile does not represent a fundamental physical entity, but rather an effective geometrical approximation of the deeper dynamics of the GDS field.

7.3 MOND (Modified Newtonian Dynamics)

MOND introduces a modification in the low-acceleration limit:

$$a \ll a_0. \quad (33)$$

In this regime, gravitational enhancement occurs.
Within GDS:

$$\rho \ll \rho_c \quad \Rightarrow \quad \Gamma(\rho) > 1. \quad (34)$$

This means:

- MOND corresponds to the low-density limit of GDS,
- GDS provides a more general framework without the need to introduce a critical acceleration.

7.4 Comparison of Approaches

Model	Mechanism	Parameter	Interpretation
Newton	Linear gravity	–	Insufficient
MOND	Acceleration modification	a_0	Phenomenological
NFW	Added matter	ρ_0, r_s	Empirical
GDS	Dynamic field	A, ρ_c, n	Generative

7.5 Key Difference

The essential difference between the approaches can be formulated as follows:

- Newton: gravity is fixed
- NFW: new matter is added
- MOND: the law is changed
- GDS: the field response changes

7.6 Interpretation of Results

Benchmark results show:

1. Newton fails without an additional mechanism
2. NFW provides a precise fit, but without a physical explanation
3. MOND captures part of the effect, but is not general
4. GDS reproduces all of these regimes within a single model

7.7 Conclusion of the Section

The gravitationally dynamic field represents a unifying framework in which Newtonian dynamics, MOND, and the NFW profile appear as different limiting or effective descriptions of the same phenomenon.

This result suggests that the differences between current approaches may be more a matter of interpretation than of fundamental physics.

7.8 Clarification of the Statement

This statement stands on three independent pillars:

1. **Numerical Reproduction of Data**

The GDS model achieves comparable or better fit quality than standard halo-containing models.

2. **Generativity of Halo Structure**

The effective halo can be derived directly from GDS without introducing new matter.

3. **Compatibility with Existing Models**

Newton, MOND, and NFW appear as limiting or effective cases of GDS.

7.9 Interpretation

The main statement does not deny the existence of observed phenomena (flat rotation curves, halo effects), but offers an alternative interpretation of their origin.

- The observed effect remains
- The interpretation of its cause changes

7.10 Range of Validity

The statement applies to:

- galactic rotation curves
- baryon-dominant galactic dynamics

and does not make direct conclusions about:

- cosmological scales (CMB, large-scale structure)
- particle physics

7.11 Conclusion of the Section

The main statement summarizes the core of this work:

The GDS model provides an alternative and consistent explanation of galactic dynamics without the need to introduce dark matter as a separate component.

8 Conclusion

In this work, we performed a systematic comparison of three approaches to describing galactic rotation curves: Newtonian dynamics, the NFW halo model, and the gravitationally dynamic field (GDS).

The results can be summarized into several key points:

- The Newtonian model without additional effects systematically fails.
- The Newton + NFW model provides a precise fit to the data, but at the cost of introducing an additional empirically motivated component.
- The GDS model achieves comparable or better agreement with observations without the need to introduce new matter.

The fundamental result of this work is the demonstration that:

The halo effect, traditionally attributed to dark matter, can arise as a natural consequence of the dynamic response of the gravitational field to baryonic structure.

Furthermore, it was numerically and analytically demonstrated that:

- the GDS model generates effective halo structures,

- these structures are compatible with the NFW profile,
- the NFW profile can be understood as an approximation of deeper dynamics.

This result leads to a fundamental interpretation:

It is not necessary to introduce a separate dark matter component if the same phenomena can be reproduced by the dynamics of the gravitational field.

From the point of view of physics, this means that the problem of galactic dynamics may not be a question of missing matter, but a question of an incomplete description of gravitational interaction.

8.1 Newtonian Dynamics

Newtonian gravity represents the fundamental limit:

$$v^2(r) = \frac{GM(r)}{r}. \quad (35)$$

This relation is exactly reproduced in the high-density limit:

$$\rho \gg \rho_c \quad \Rightarrow \quad \Gamma(\rho) \rightarrow 1. \quad (36)$$

Newtonian dynamics is therefore not invalidated, but represents a special case of GDS.

8.2 NFW Profile and Λ CDM

The standard cosmological model introduces dark matter through a halo:

$$v^2(r) = v_{\text{bar}}^2(r) + v_{\text{halo}}^2(r). \quad (37)$$

The NFW profile arises as an empirical result of simulations.

Within the GDS framework:

- the NFW profile can be reproduced without introducing new matter,
- its parameters are derivable from the baryonic structure.

This implies:

The NFW halo can be interpreted as an effective description of a dynamical gravitational field. The NFW profile does not represent a fundamental physical entity, but an effective geometrical approximation of deeper GDS field dynamics.

8.3 MOND (Modified Newtonian Dynamics)

MOND introduces a modification in the low-acceleration regime:

$$a \ll a_0. \quad (38)$$

In this regime, gravity is enhanced.

Within GDS:

$$\rho \ll \rho_c \quad \Rightarrow \quad \Gamma(\rho) > 1. \quad (39)$$

This means:

- MOND corresponds to the low-density limit of GDS,
- GDS provides a more general framework without introducing a critical acceleration scale.

8.4 Comparison of Approaches

Model	Mechanism	Parameter	Interpretation
Newton	Linear gravity	–	Insufficient
MOND	Modified acceleration	a_0	Phenomenological
NFW	Added matter	ρ_0, r_s	Empirical
GDS	Dynamic field	A, ρ_c, n	Generative

8.5 Key Difference

The fundamental difference between the approaches can be formulated as:

- Newton: gravity is fixed
- NFW: new matter is added
- MOND: the law is modified
- GDS: the field response is modified

8.6 Interpretation of Results

The benchmark results show:

1. Newton fails without an additional mechanism
2. NFW provides an accurate fit, but without a physical explanation
3. MOND captures part of the effect, but is not general
4. GDS reproduces all these regimes within a single model

8.7 Conclusion of the Section

The gravitationally dynamic field represents a unifying framework in which Newtonian dynamics, MOND, and the NFW profile appear as different limiting or effective descriptions of the same phenomenon.

This suggests that the differences between current approaches may be a matter of interpretation rather than fundamental physics.

9 Main Statement of the Paper

Based on numerical simulations, statistical benchmarking, and analytical reconstruction of the effective halo, we formulate the following central statement:

Effects traditionally attributed to dark matter in galactic rotation curves can be reproduced by a gravitationally dynamic field that modifies the effective gravitational response as a function of local baryonic density, without the need to introduce a separate dark matter component.

9.1 Clarification of the Statement

This statement rests on three independent pillars:

1. **Numerical reproduction of data**
The GDS model achieves comparable or better fit quality than standard halo-based models.
2. **Generativity of halo structure**
The effective halo can be derived directly from GDS without introducing new matter.
3. **Compatibility with existing models**
Newton, MOND, and NFW appear as limiting or effective cases of GDS.

9.2 Interpretation

The main statement does not deny the existence of observed phenomena (flat rotation curves, halo effects), but provides an alternative interpretation of their origin.

- The observed effect remains
- The interpretation of its cause changes

9.3 Scope of Validity

The statement applies to:

- galactic rotation curves
- baryon-dominated galaxy dynamics

and does not make direct claims about:

- cosmological scales (CMB, large-scale structure)
- particle physics

9.4 Conclusion of the Section

The GDS model provides an alternative and consistent explanation of galactic dynamics without the need to introduce dark matter as a separate component.

10 Conclusion

In this work, we performed a systematic comparison of three approaches to describing galactic rotation curves: Newtonian dynamics, the Newton + NFW halo model, and the gravitationally dynamic field (GDS).

The results can be summarized as follows:

- The Newtonian model without additional effects systematically fails.
- The Newton + NFW model provides an accurate fit, but at the cost of introducing an additional empirically motivated component.
- The GDS model achieves comparable or better agreement with observations without introducing new matter.

A key result of this work is the demonstration that:

The halo effect, traditionally attributed to dark matter, can arise as a natural consequence of the dynamical response of the gravitational field to baryonic structure.

Furthermore, it has been shown both numerically and analytically that:

- the GDS model generates effective halo structures,
- these structures are compatible with the NFW profile,
- the NFW profile can be interpreted as an approximation of deeper dynamics.

This leads to a fundamental interpretation:

It is not necessary to introduce a separate dark matter component if the same phenomena can be reproduced by gravitational field dynamics.

From a physical standpoint, this suggests that the problem of galactic dynamics may not be a question of missing matter, but rather of an incomplete description of gravitational interaction.

11 Author's Perspective

This work was developed outside the traditional academic environment. It is the result of a practical approach combining numerical testing, physical intuition, and strict logical reasoning.

Gravity is present in every aspect of reality. It is therefore natural to seek its deeper dynamics before introducing new, unobserved entities.

This work is not only a technical comparison of models. It is the result of a long-term effort to understand whether so-called dark phenomena represent an independent physical reality, or merely the consequence of an incomplete description of gravity.

Based on the performed calculations, simulations, and logical analysis, I arrive at the following conclusion:

Dark matter and dark energy represent a misinterpreted gravitationally dynamic field, which manifests itself as an effective halo structure depending on baryonic density.

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Code and Data Availability

All scripts, numerical tests, and generated results are reproducible based on the methodology described in this work.

The code includes:

- modeling of rotation curves (disk + gas),
- implementation of the GDS field,
- optimization benchmark (Newton, NFW, GDS),
- statistical evaluation (AIC, BIC).

Computational Environment

All computations and simulations presented in this work were performed on a personal device:

- Device: LAPTOP-EJVF49MH (Lenovo / IBM)
- Processor: AMD Ryzen 7 5700U (1.80 GHz)
- Memory: 16 GB RAM
- Storage: SSD
- Operating system: Linux-based environment

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